QML Lab • research project 13

Learning from Classical Shadows

Predicting many properties of a quantum state from few randomised measurements.

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Quadrant	QC (quantum data or quantum algorithm, classical learner)	**Track**	Quantum data
Difficulty	Hard	**Tools**	PennyLane, NumPy, scikit-learn
Builds on	Modules 2 and 8 of the QML Lab	**Workspace**	`qml_projects/classical-shadows/`

Summary. Full quantum state tomography needs a number of measurements that grows exponentially with the number of qubits. Classical shadows replace it with randomised single-qubit measurements and simple classical post-processing, and predict many properties at once with far fewer measurements. This project measures how estimation error scales, compares shadows with tomography, and uses shadow-based features to train a classical phase classifier, linking it to the QCNN of Project 12.

1 Background and key concepts

1.1 Randomised measurements

Randomised measurement protocols apply random unitaries before measuring and learn from the statistics of the outcomes (Elben et al., 2023). In the classical-shadow protocol of Huang et al. (2020), each round applies a random single-qubit Clifford U_j to every qubit (a random choice of the X, Y or Z basis), measures, and stores the snapshot

$$\hat{\rho} = \bigotimes_{j=1}^{n} \Big(3\, U_j^\dagger \, |\hat{b}_j\rangle\langle\hat{b}_j| \, U_j - I \Big), \tag{1}$$

which is an unbiased estimator of ρ: $\mathbb{E}[\hat{\rho}] = \rho$.

1.2 Sample complexity

Expectation values $o_i = \mathrm{Tr}(O_i \rho)$ are estimated by median-of-means over snapshots. To predict M observables to accuracy ϵ with high probability, it suffices to take

$$N_{\mathrm{snap}} = O\Big(\frac{\log M}{\epsilon^2} \max_i \|O_i\|^2_{\mathrm{shadow}} \Big), \qquad \|O\|^2_{\mathrm{shadow}} \le 3^k \|O\|^2_\infty \tag{2}$$

for k-local observables with Pauli measurements. Local properties are cheap; global ones are not. Non-linear quantities such as the purity $\mathrm{Tr}\,\rho_A^2$ and the second Rényi entropy $S_2(\rho_A) = -\log \mathrm{Tr}\,\rho_A^2$ follow from pairs of independent snapshots.

1.3 Machine learning on shadows

Huang et al. (2022) proved that classical machine learning trained on classical-shadow data can efficiently predict ground-state properties and classify phases of matter. Neural-network tomography (Torlai et al., 2018) is an alternative that learns a model of the state from measurements. Both are **QC**: the data comes from quantum states, the learning is classical.

2 Research questions

The project is open-ended: these questions are the starting point, not the finish line. Add new ones to the README and the journal as they come up.

RQ1 How does the estimation error of local observables and of subsystem entropy scale with the number of snapshots?

RQ2 Which observables are cheap to estimate and which need prohibitively many snapshots?

RQ3 Can a classical model trained on shadow features classify Ising phases as well as the QCNN of Project 12, for the same number of state copies?

3 Method

3.1 Data

GHZ states, random product states and Ising ground states (shared with Project 12) on 4 to 10 qubits; 10^2 to 10^4 snapshots per state, generated with PennyLane's classical-shadow measurement.

3.2 Models

Shadow estimators for $\langle X_i \rangle$, $\langle Z_i Z_{i+1} \rangle$, the fidelity with a target state and S_2 of half the chain; full tomography by linear inversion for $n \leq 4$; a logistic regression or small MLP on shadow-estimated correlations for phase classification.

3.3 Baselines

Exact values from the state vector; full tomography with the same number of measurements; direct measurement in a fixed basis for each observable.

3.4 Metrics

- Absolute estimation error versus number of snapshots (log–log).
- Snapshots needed to reach $\epsilon = 0.05$.
- Phase classification accuracy versus number of state copies.

4 Planned experiments

Each experiment becomes one `[[experiments]]` entry in the workspace, so its runs, parameters and metrics are stored and compared in the portal.

ID	Experiment	What varies	Main metrics
E1	Local observables	$N_{\text{snap}} = 10^2..10^4$	error scaling
E2	Entanglement	S_2 of half chain	error vs N_{snap}
E3	Tomography comparison	$n \leq 4$, equal budget	error
E4	Phase classifier	shadow features of Ising states	accuracy vs copies
E5	Readout noise	bit-flip probability	bias

Table 1: Experiment plan. Results go into the journal and the results log (section 9).

5 Running it in the QML Lab

The workspace lives in `qml_projects/classical-shadows/`. Scripts import three helpers from `qml_lab/plots.py`:

- `params()` reads the parameters from the portal's run form (defaults when run by hand);
- `save()` stores figures with the run;
- `log_metrics()` records numbers that appear in the run table.

```
qml_projects/classical-shadows/
  project.toml          metadata + experiments
  README.md             goal, open questions, ideas
  experiments/          runnable scripts
  docs/main.tex         this document
```

First experiment: This experiment is *planned*: write the script, add the block below to `project.toml`, then run `python manage.py sync_projects`.

```
[[experiments]]
slug = "shadow-scaling"
title = "Error scaling of classical-shadow estimates"
script = "experiments/e01_shadow_scaling.py"
figure = "error_scaling.png"

[experiments.params]
n_qubits = 6
n_snapshots = 1000
state = "ising"
field = 1.0
seed = 0
```

```
# from the portal folder: pick up changes to project.toml
python manage.py sync_projects
cd qml_projects/classical-shadows
# run by hand with the default parameters
PYTHONPATH=../../qml_lab python experiments/e01_shadow_scaling.py
```

6 Quality bar

- A classical baseline with a similar number of trainable parameters, tuned with the same care (Bowles et al., 2024).
- At least five random seeds per configuration, reported as mean $\pm$ standard deviation.
- How results change with the number of qubits.
- At least one run under a realistic noise model.
- Fit and report the error-scaling exponent; it should be close to $-1/2$.

7 Open directions

- Derandomised shadows for a fixed set of observables.
- Shadows from real hardware, including readout mitigation.
- Shared data set with Project 12 for a direct **QC** versus **QQ** comparison.

8 Risks and limitations

- Global observables need exponentially many snapshots; choose observables with this in mind.
- Readout errors bias estimates unless mitigated.
- Median-of-means parameters affect the error; keep them fixed across comparisons.

9 Results log

Summarise findings here as they become stable; day-to-day notes belong in the portal journal.

Date	Run	Finding	Next step

References

Bowles, J., Ahmed, S. and Schuld, M. (2024) 'Better than classical? The subtle art of benchmarking quantum machine learning models', Preprint, arXiv:2403.07059. Available at: https://arxiv.org/abs/2403.07059.

Elben, A., Flammia, S. T., Huang, H.-Y., Kueng, R., Preskill, J., Vermersch, B. and Zoller, P. (2023) 'The randomized measurement toolbox', *Nature Reviews Physics*, 5, pp. 9–24. Available at: https://doi.org/10.1038/s42254-022-00535-2.

Huang, H.-Y., Kueng, R. and Preskill, J. (2020) 'Predicting many properties of a quantum system from very few measurements', *Nature Physics*, 16, pp. 1050–1057. Available at: https://doi.org/10.1038/s41567-020-0932-7.

Huang, H.-Y., Kueng, R., Torlai, G., Albert, V. V. and Preskill, J. (2022) 'Provably efficient machine learning for quantum many-body problems', *Science*, 377(6613), eabk3333. Available at: https://doi.org/10.1126/science.abk3333.

Torlai, G., Mazzola, G., Carrasquilla, J., Troyer, M., Melko, R. and Carleo, G. (2018) 'Neural-network quantum state tomography', *Nature Physics*, 14, pp. 447–450. Available at: https://doi.org/10.1038/s41567-018-0048-5.